مدیریت ورود و حجم مصرفی کاربران سرورمجازی

۱- اضافه کردن پورت های مجاز

```
  GNU nano 7.2                                  /etc/ssh/sshd_config

# This is the sshd server system-wide configuration file.  See
# sshd_config(5) for more information.

# This sshd was compiled with PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games

# The strategy used for options in the default sshd_config shipped with
# OpenSSH is to specify options with their default value where
# possible, but leave them commented.  Uncommented options override the
# default value.

Include /etc/ssh/sshd_config.d/*.conf

Port 22
Port 4280
Port 3290
Port 6040
Port 3654
Port 9546
Port 1254

#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::
```

۲- اضافه کردن قواعد ورودی و خروجی برای هر کاربر

```
root@test:/home/ubuntu# iptables -I INPUT -p tcp --dport 4280 -j ACCEPT
root@test:/home/ubuntu# iptables -I INPUT -p tcp --dport 3290 -j ACCEPT
root@test:/home/ubuntu# iptables -I INPUT -p tcp --dport 6040 -j ACCEPT
root@test:/home/ubuntu# iptables -I INPUT -p tcp --dport 3654 -j ACCEPT
root@test:/home/ubuntu# iptables -I INPUT -p tcp --dport 9546 -j ACCEPT
root@test:/home/ubuntu# iptables -I INPUT -p tcp --dport 1254 -j ACCEPT
root@test:/home/ubuntu#  iptables -I OUTPUT -p tcp --sport 4280 -j ACCEPT
root@test:/home/ubuntu#  iptables -I OUTPUT -p tcp --sport 3290 -j ACCEPT
root@test:/home/ubuntu#  iptables -I OUTPUT -p tcp --sport 6040 -j ACCEPT
root@test:/home/ubuntu#  iptables -I OUTPUT -p tcp --sport 3654 -j ACCEPT
root@test:/home/ubuntu#  iptables -I OUTPUT -p tcp --sport 9546 -j ACCEPT
root@test:/home/ubuntu#  iptables -I OUTPUT -p tcp --sport 1254 -j ACCEPT
root@test:/home/ubuntu# iptables -L -v -n
Chain INPUT (policy ACCEPT 119K packets, 248M bytes)
 pkts bytes target     prot opt in     out     source               destination
    0     0 ACCEPT     6    --  *      *       0.0.0.0/0            0.0.0.0/0            tcp dpt:1254
    0     0 ACCEPT     6    --  *      *       0.0.0.0/0            0.0.0.0/0            tcp dpt:9546
    0     0 ACCEPT     6    --  *      *       0.0.0.0/0            0.0.0.0/0            tcp dpt:3654
    0     0 ACCEPT     6    --  *      *       0.0.0.0/0            0.0.0.0/0            tcp dpt:6040
    0     0 ACCEPT     6    --  *      *       0.0.0.0/0            0.0.0.0/0            tcp dpt:3290
    0     0 ACCEPT     6    --  *      *       0.0.0.0/0            0.0.0.0/0            tcp dpt:4280
13779 1015K ACCEPT     6    --  *      *       0.0.0.0/0            0.0.0.0/0            tcp dpt:22

Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain OUTPUT (policy ACCEPT 98083 packets, 5555K bytes)
 pkts bytes target     prot opt in     out     source               destination
    0     0 ACCEPT     6    --  *      *       0.0.0.0/0            0.0.0.0/0            tcp spt:1254
    0     0 ACCEPT     6    --  *      *       0.0.0.0/0            0.0.0.0/0            tcp spt:9546
    0     0 ACCEPT     6    --  *      *       0.0.0.0/0            0.0.0.0/0            tcp spt:3654
    0     0 ACCEPT     6    --  *      *       0.0.0.0/0            0.0.0.0/0            tcp spt:6040
    0     0 ACCEPT     6    --  *      *       0.0.0.0/0            0.0.0.0/0            tcp spt:3290
    0     0 ACCEPT     6    --  *      *       0.0.0.0/0            0.0.0.0/0            tcp spt:4280
11219 1490K ACCEPT     6    --  *      *       0.0.0.0/0            0.0.0.0/0            tcp spt:22
```

۳- نصب پکیج iptables-persistent

```
root@test:/home/ubuntu# apt install iptables-persistent
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  netfilter-persistent
The following packages will be REMOVED:
  ufw
The following NEW packages will be installed:
  iptables-persistent netfilter-persistent
0 upgraded, 2 newly installed, 1 to remove and 186 not upgraded.
Need to get 14.3 kB of archives.
```

۴ – ریستارت سرور و اجرای دستورات زیر

```
root@test:/home/ubuntu# ss -tuln
Netid      State      Recv-Q     Send-Q              Local Address:Port          Peer Address:Port     Process
udp        UNCONN     0          0                      127.0.0.54:53                0.0.0.0:*
udp        UNCONN     0          0                   127.0.0.53%lo:53                0.0.0.0:*
udp        UNCONN     0          0              172.20.89.242%ens3:68                0.0.0.0:*
tcp        LISTEN     0          4096                127.0.0.53%lo:53                0.0.0.0:*
tcp        LISTEN     0          128                       0.0.0.0:9546              0.0.0.0:*
tcp        LISTEN     0          128                       0.0.0.0:1254              0.0.0.0:*
tcp        LISTEN     0          128                       0.0.0.0:3290              0.0.0.0:*
tcp        LISTEN     0          128                       0.0.0.0:6040              0.0.0.0:*
tcp        LISTEN     0          4096                   127.0.0.54:53                0.0.0.0:*
tcp        LISTEN     0          128                       0.0.0.0:3654              0.0.0.0:*
tcp        LISTEN     0          128                       0.0.0.0:4280              0.0.0.0:*
tcp        LISTEN     0          128                       0.0.0.0:22                0.0.0.0:*
tcp        LISTEN     0          128                          [::]:9546                 [::]:*
tcp        LISTEN     0          128                          [::]:1254                 [::]:*
tcp        LISTEN     0          128                          [::]:3290                 [::]:*
tcp        LISTEN     0          128                          [::]:6040                 [::]:*
tcp        LISTEN     0          128                          [::]:3654                 [::]:*
tcp        LISTEN     0          128                          [::]:4280                 [::]:*
tcp        LISTEN     0          128                          [::]:22                   [::]:*
```

```
root@srv8484478595:~# iptables -L -v -n
Chain INPUT (policy ACCEPT 225 packets, 17531 bytes)
 pkts bytes target     prot opt in     out     source               destination
    0     0 ACCEPT     6    --  *      *       0.0.0.0/0            0.0.0.0/0            tcp dpt:1254
    0     0 ACCEPT     6    --  *      *       0.0.0.0/0            0.0.0.0/0            tcp dpt:9546
    0     0 ACCEPT     6    --  *      *       0.0.0.0/0            0.0.0.0/0            tcp dpt:3654
    0     0 ACCEPT     6    --  *      *       0.0.0.0/0            0.0.0.0/0            tcp dpt:6040
    0     0 ACCEPT     6    --  *      *       0.0.0.0/0            0.0.0.0/0            tcp dpt:3290
    0     0 ACCEPT     6    --  *      *       0.0.0.0/0            0.0.0.0/0            tcp dpt:4280

Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain OUTPUT (policy ACCEPT 196 packets, 21674 bytes)
 pkts bytes target     prot opt in     out     source               destination
    0     0 ACCEPT     6    --  *      *       0.0.0.0/0            0.0.0.0/0            tcp spt:1254
    0     0 ACCEPT     6    --  *      *       0.0.0.0/0            0.0.0.0/0            tcp spt:9546
    0     0 ACCEPT     6    --  *      *       0.0.0.0/0            0.0.0.0/0            tcp spt:3654
    0     0 ACCEPT     6    --  *      *       0.0.0.0/0            0.0.0.0/0            tcp spt:6040
    0     0 ACCEPT     6    --  *      *       0.0.0.0/0            0.0.0.0/0            tcp spt:3290
    0     0 ACCEPT     6    --  *      *       0.0.0.0/0            0.0.0.0/0            tcp spt:4280
```

۵ – ایجاد فایل های block_duplicate_login.sh و iptables-save-restore.sh وقابل اجرایی کردن آن ها

```
  GNU nano 7.2                    /usr/local/bin/block_duplicate_login.sh *
#!/bin/bash


PORT_FILE="/root/monitor_users/ports.txt"

[ "$PAM_USER" = "root" ] && exit 0

allowed_port=$(grep "^$PAM_USER:" "$PORT_FILE" | cut -d':' -f2)

if [ -z "$allowed_port" ]; then

        exit 1
fi


session_count=$(ps aux | awk -v u="$PAM_USER" '$1 == u && $0 ~ ("sshd: " u)' | wc -l)

if [ "$session_count" -ge 2 ]; then
    exit 1
fi

pids=$(ps aux | awk -v u="$PAM_USER" '$1 == "sshd" && $0 ~ ("sshd: " u) { print $2 }')

for pid in $pids; do
    port=$(sudo lsof -Panp "$pid" -i | awk '/TCP/ && /ESTABLISHED/ { split($9, a, ":"); split(a[2], b, "->"); print b[1]

    if [ "$port" != "$allowed_port" ]; then
        exit 1
    fi
done

exit 0
```

```
  GNU nano 7.2                    /usr/local/bin/iptables-save-restore.sh
#!/bin/bash

RULES_FILE="/etc/iptables/rules.v4"

case "$1" in
  save)
    iptables-save -c > "$RULES_FILE"
    ;;
  restore)
    if [[ -f "$RULES_FILE" ]]; then

      iptables-restore -c < "$RULES_FILE"
    else
      exit 1
    fi
    ;;
  *)
    exit 1
    ;;
esac
```

```
root@DESKTOP-8S2U0EG /u/l/bin# chmod +x /usr/local/bin/block_duplicate_login.sh
root@DESKTOP-8S2U0EG /u/l/bin# chmod +x /usr/local/bin/iptables-save-restore.sh
root@DESKTOP-8S2U0EG /u/l/bin# ls -la
total 16
drwxr-xr-x  2 root root 4096 Jul 24 16:27 ./
drwxr-xr-x 10 root root 4096 Feb 15 11:39 ../
-rwxr-xr-x  1 root root  664 Jul 23 22:40 block_duplicate_login.sh*
-rwxr-xr-x  1 root root  270 Jul 23 19:48 iptables-save-restore.sh*
lrwxrwxrwx  1 root root   55 Jul 24 10:23 kubectl -> /mnt/wsl/docker-desktop/cli-tools/usr/local/bin/kubectl*
```

۶ - زدن دستور crontab – e و اضافه کردن دو خط زیر برای اجرای خودکارآن ها

```
  GNU nano 7.2                                  /tmp/crontab.DgRSZE/crontab *
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h  dom mon dow   command
* * * * * /root/monitor_users/venv/bin/python3  /root/monitor_users/monitor_users_app.py
@reboot /usr/local/bin/iptables-save-restore.sh restore
```

۷- ایجاد فایل زیر برای ذخیره ترافیک ورودی و خروجی پورت ها هنگام افتادن سرور

```
  GNU nano 7.2                    /etc/systemd/system/iptables-save-before-shutdown.service
[Unit]
Description=Save iptables rules with counters before shutdown/reboot
DefaultDependencies=no
Before=shutdown.target reboot.target halt.target

[Service]
Type=oneshot
ExecStart=/usr/local/bin/iptables-save-restore.sh save
RemainAfterExit=true

[Install]
WantedBy=halt.target reboot.target shutdown.target
```

۸- اضافه کردن خط زیر به ابتدای فایل

```
  GNU nano 7.2                                        /etc/pam.d/sshd *
auth required pam_exec.so /usr/local/bin/block_duplicate_login.sh


# PAM configuration for the Secure Shell service

# Standard Un*x authentication.
@include common-auth

# Disallow non-root logins when /etc/nologin exists.
account    required     pam_nologin.so
```

۹- اضافه کردن کاربران به سرور

```
root@DESKTOP-8S2UOEG /m/c/U/pc# adduser  ashkan --shell /usr/sbin/nologin
info: Adding user `ashkan' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `ashkan' (1002) ...
info: Adding new user `ashkan' (1002) with group `ashkan (1002)' ...
info: Creating home directory `/home/ashkan' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for ashkan
Enter the new value, or press ENTER for the default
        Full Name []:
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n]
info: Adding new user `ashkan' to supplemental / extra groups `users' ...
info: Adding user `ashkan' to group `users' ...
root@DESKTOP-8S2UOEG /m/c/U/pc# adduser  mohammadreza --shell /usr/sbin/nologin
info: Adding user `mohammadreza' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `mohammadreza' (1003) ...
info: Adding new user `mohammadreza' (1003) with group `mohammadreza (1003)' ...
info: Creating home directory `/home/mohammadreza' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for mohammadreza
Enter the new value, or press ENTER for the default
        Full Name []:
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n]
info: Adding new user `mohammadreza' to supplemental / extra groups `users' ...
info: Adding user `mohammadreza' to group `users' ...
```

۱۰- clone کردن پروژه از گیتهاب در مسیر root/ و ایجاد محیط مجازی

```
root@DESKTOP-8S2UOEG:~/monitor_users# apt install python3-venv
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
python3-venv is already the newest version (3.12.3-0ubuntu2).
0 upgraded, 0 newly installed, 0 to remove and 63 not upgraded.
root@DESKTOP-8S2UOEG:~/monitor_users# python3 -m venv venv
root@DESKTOP-8S2UOEG:~/monitor_users# source venv/bin/activate
(venv) root@DESKTOP-8S2UOEG:~/monitor_users#
```

۱۱- ایجاد پایگاه داده برای نگهداری اطلاعات کاربران با استفاده از docker

```
docker run -d --name mysql-db -e MYSQL_ROOT_PASSWORD=aBc.123456 -e MYSQL_DATABASE=monitor_users_db -v ./db_content:/var/li
b/mysql -p 3306:3306 mysql:5.7
Unable to find image 'mysql:5.7' locally
5.7: Pulling from library/mysql
20e4dcae4c69: Pull complete
1c56c3d4ce74: Pull complete
e9f03a1c24ce: Pull complete
68c3898c2015: Pull complete
6b95a940e7b6: Pull complete
90986bb8de6e: Pull complete
ae71319cb779: Pull complete
ffc89e9dfd88: Pull complete
43d05e938198: Pull complete
064b2d298fba: Pull complete
df9a4d85569b: Pull complete
Digest: sha256:4bc6bc963e6d8443453676cae56536f4b8156d78bae03c0145cbe47c2aad73bb
Status: Downloaded newer image for mysql:5.7
9493bf85b9b1fb8754dd291b21e82267cee7a324dfb137895488e500df2f0eba
root@srv8484478595:~/monitor_users# docker ps
CONTAINER ID   IMAGE       COMMAND                  CREATED          STATUS          PORTS
                    NAMES
9493bf85b9b1   mysql:5.7   "docker-entrypoint.s…"   12 seconds ago   Up 12 seconds   0.0.0.0:3306->3306/tcp, [::]:3306->33
06/tcp, 33060/tcp   mysql-db
```

۱۲- ورود به پایگاده داده و ایجاد جدول کاربران و درج اطلاعات آن ها در جدول

```
root@DESKTOP-8S2UOEG ~/monitor_users (main)# docker exec -it mysql-db /bin/bash
bash-5.1# mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 457
Server version: 9.3.0 MySQL Community Server - GPL

Copyright (c) 2000, 2025, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> use monitor_users_db;
Database changed
mysql> CREATE TABLE users (
    ->     id VARCHAR(30),
    ->     password VARCHAR(30),
    ->     chat_id VARCHAR(10),
    ->     port VARCHAR(10),
    ->     download_volume VARCHAR(50) ,
    ->     upload_volume VARCHAR(50),
    ->     PRIMARY KEY (id)
    -> );
Query OK, 0 rows affected (0.078 sec)
```

```
mysql> insert into users values("ashkan","4280","7858487903", "4280" , "0" , "0");
Query OK, 1 row affected (0.035 sec)

mysql> insert into users values("mohammadreza","3290","7980195182", "3290" , "0" , "0");
Query OK, 1 row affected (0.038 sec)
```

۱۳ – نصب نیازمندی های پوشه requirements و سپس اجرای بات

```
  GNU nano 7.2                    /etc/systemd/system/telegrambot.service
[Unit]
Description=Telegram Bot Service
After=network.target mysql.service

[Service]
Type=simple
WorkingDirectory=/root/monitor_users
ExecStart=/root/monitor_users/venv/bin/python3 bot.py
Restart=always
User=root
Environment=PYTHONUNBUFFERED=1

[Install]
WantedBy=multi-user.target
```

```
root@DESKTOP-8S2UOEG ~# systemctl daemon-reexec
root@DESKTOP-8S2UOEG ~# systemctl daemon-reload
root@DESKTOP-8S2UOEG ~# systemctl enable telegrambot.service
Created symlink /etc/systemd/system/multi-user.target.wants/telegrambot.service → /etc/systemd/system/telegrambot.servic
e.
root@DESKTOP-8S2UOEG ~# systemctl is-enabled telegrambot.service
enabled
root@DESKTOP-8S2UOEG ~# systemctl start telegrambot.service
```

```
root@DESKTOP-8S2UOEG ~# systemctl status telegrambot.service
● telegrambot.service - Telegram Bot Service
     Loaded: loaded (/etc/systemd/system/telegrambot.service; enabled; preset: enabled)
     Active: active (running) since Thu 2025-07-24 18:13:39 +0330; 1min 10s ago
   Main PID: 29046 (python3)
      Tasks: 2 (limit: 9294)
     Memory: 30.4M (peak: 31.1M)
        CPU: 230ms
     CGroup: /system.slice/telegrambot.service
             └─29046 /root/monitor_users/venv/bin/python3 bot.py

Jul 24 18:13:39 DESKTOP-8S2UOEG systemd[1]: telegrambot.service: Scheduled restart job, restart counter is at 1.
Jul 24 18:13:39 DESKTOP-8S2UOEG systemd[1]: Started telegrambot.service - Telegram Bot Service.
root@DESKTOP-8S2UOEG ~#
```